LOCAL MULTIMODAL AGENTIC RAG • CALIBRATED EVALUATION

Locally Measured, Holdout-Validated Multimodal RAG with Hybrid pgvector Search and 7-Layer Verification.

A framework-free Python architecture (<1,100 modular core lines) combining **Docling + Chandra OCR 2** for adaptive page ingestion, **PostgreSQL pgvector + Full-Text RRF** and **BGE Cross-Encoder Reranking** for retrieval, **Qwen3-VL-4B** for visual-textual grounded streaming answers, and a **Cohen's κ calibrated evaluation suite**.

Explore Final Holdout & Phases · Tech Slide Deck (PDF) · Non-Tech Slide Deck (PDF) · Model Card · NIST AI RMF · View Failure Post-Mortems

0.98 / 0.94	**0.91 / 0.95**	**κ = 0.79 – 1.00**
Holdout R@6 & Faithfulness	Final Correctness & Gold Page Cited	Calibrated Holdout Judge Agreement

rag.py — Agentic Multi-Hop & VLM Grounding

```
# 1. Standalone query rewrite + up to 3 agentic search rounds
Query: "Which two mirror shapes are used in the grazing incidence X-ray telescope?"
Round 1: Hybrid RRF (40 candidates) → BGE Reranker (Top 6 of 1,164 chunks)
Decision: ENOUGH (top rerank_score = 0.994, doc3.pdf p.156–157)

# 2. VLM receives top text chunks + 1966 scanned schematic image (896px)
Visual Input: [doc3.pdf page 157 (Chandra OCR + PIL Image)]
Answer: "The two mirror shapes used in the grazing incidence X-ray
telescope are a paraboloid and a hyperboloid [1]."
✓ Grounded Citation: doc3.pdf p.157 [1] • Query Rerank Score: 0.994
```

OVERVIEW

What this project demonstrates

Designed without heavy orchestration wrappers (<1,100 lines of modular core Python), every component is explicitly measurable, protected against untrusted document inputs, and validated on real scientific and technical PDFs.

`indexer.py • file_processing.py`

Adaptive Hybrid PDF Ingestion

Measures per-page largest object area (`image_coverage` via `pypdfium2`). Born-digital text and tables are extracted in seconds via **Docling** (`text_layer`), while scanned pages or pages with large diagrams (`coverage ≥ 0.30`) automatically route to **Chandra OCR 2** (`datalab-to/chandra-ocr-2`).

`db.py • PostgreSQL + pgvector`

SQL-Native Hybrid Search (RRF) + Reranking

Combines 1024-dim **BAAI/bge-m3** cosine similarity (HNSW index) with PostgreSQL TSVECTOR full-text search (GIN index) inside a single SQL CTE using **Reciprocal Rank Fusion (k=60)**, then re-scores top-40 candidates with **BAAI/bge-reranker-v2-m3**.

`rag.py • model.py`

Agentic Multi-Hop Search & VLM Grounding

Rewrites conversational follow-ups into standalone queries, evaluates whether retrieved chunks are sufficient (`ENOUGH` vs. `SEARCH: <new query>` up to 3 rounds), and passes both text excerpts and rendered page PNGs to **Qwen3-VL-4B-Instruct** within a strict 9,000-token budget.

`test_prompt_injection.py • evaluate.py`

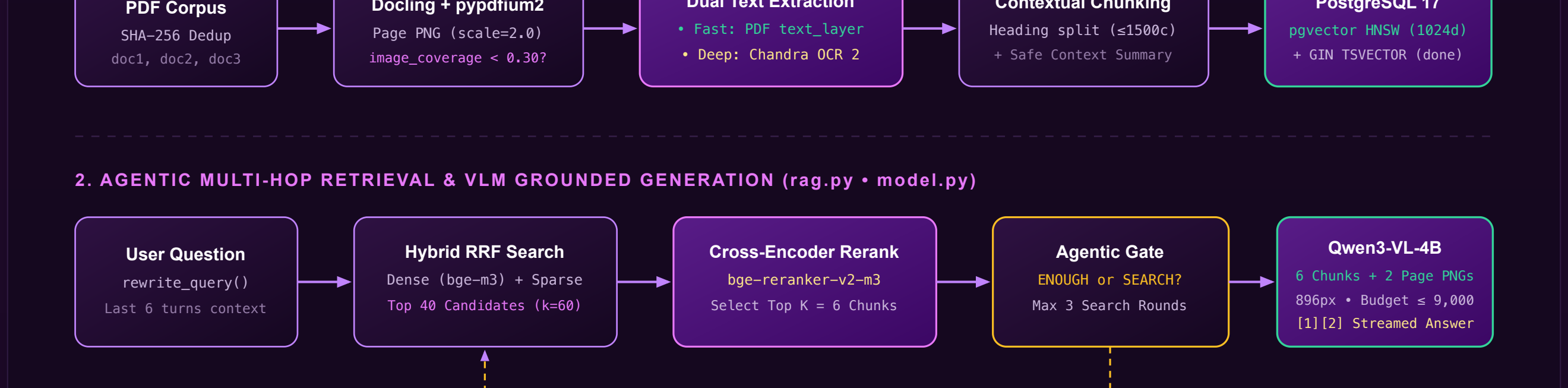
Defense-in-Depth & Hypothesis-Driven Phases

Protects against indirect prompt injection planted in PDFs (hidden white text & fake system notes) at both index time (regex canary/URL stripping on contextual summaries) and answer time. Every engineering change follows a strict *Measure → Single Change → Re-Measure* phase discipline.

ARCHITECTURE

End-to-end multimodal indexing & agentic retrieval pipeline

Models load lazily on Apple Silicon (`mps`) / CUDA with `bfloat16` precision. Chandra OCR is automatically unloaded and GPU cache is cleared once indexing completes.



1. ADAPTIVE MULTIMODAL INDEXING PIPELINE (indexer.py • file_processing.py)

- **PDF Corpus** — SHA-256 Dedup — doc1, doc2, doc3
- **Docling + pypdfium2** — Page PNG (scale=2.0) — image_coverage < 0.30?
- **Dual Text Extraction** — • Fast: PDF text_layer • Deep: Chandra OCR 2
- **Contextual Chunking** — Heading split (≤1500c) + Safe Context Summary
- **PostgreSQL 17** — pgvector HNSW (1024d) + GIN TSVECTOR (done)

2. AGENTIC MULTI-HOP RETRIEVAL & VLM GROUNDED GENERATION (rag.py • model.py)

- **User Question** — rewrite_query() — Last 6 turns context
- **Hybrid RRF Search** — Dense (bge-m3) + Sparse — Top 40 Candidates (k=60)
- **Cross-Encoder Rerank** — bge-reranker-v2-m3 — Select Top K = 6 Chunks
- **Agentic Gate** — ENOUGH or SEARCH? — Max 3 Search Rounds
- **Qwen3-VL-4B** — 6 Chunks + 2 Page PNGs — 896px • Budget ≤ 9,000 — [1][2] Streamed Answer

If missing info: emit new query & merge candidate pool

CONTINUOUS VERIFICATION & CALIBRATED EVALUATION LAYER (evaluate.py • pytest)
Recall@1/6 • MRR • Faithfulness • Citation Precision • Refusal Accuracy • TTFT & p50/p95 Latency • Stratified Cohen's κ Calibration

ARCHITECTURE DECISION RECORDS (ADR)

Key engineering decisions & explicit trade-offs

Every architectural choice was evaluated against industry-standard alternatives, documenting both the capability gained and the concrete engineering trade-off accepted.

Architecture Decision Record (ADR) Matrix — ADR-001 through ADR-005 • Production Trade-Off Analysis

ADR & COMPONENT	DECISION CHOSEN	INDUSTRY DEFAULT REJECTED	ENGINEERING RATIONALE & TRADE-OFF ACCEPTED
ADR-001 `rag.py • model.py`	`<1,100 LOC Pure Python`	LangChain / LlamaIndex wrappers	**Why:** Zero hidden prompt templates or abstraction layers; enabled custom 1,024-query `blocked_eager` attention, exact 9k token budgeting (`fit_prompt`), and streaming first-60-char refusal gating. **Trade-off:** Built custom RRF SQL, chunker, and streaming state machine from scratch.
ADR-002 `indexer.py • file_processing.py`	`Adaptive Docling + Chandra OCR 2`	PyPDF text-only or OCR-everything	**Why:** Text-only extractors fail on 1965 scanned NASA reports (`doc3`) and diagrams; OCR-everything wastes hours on born-digital PDFs. Per-page `image_coverage ≥ 0.30` routing gets the best of both worlds and frees MPS VRAM after indexing. **Trade-off:** Full 200-page scanned NASA report takes ~5.5 hrs one-time indexing, mitigated via page-level ACID checkpointing.
ADR-003 `db.py • PostgreSQL 17`	`Single-DB pgvector HNSW + FTS RRF`	Separate Vector DB (Pinecone / Qdrant) + Elastic	**Why:** Eliminates dual-write synchronization bugs. Dense 1024-d `bge-m3` cosine search and stopword-filtered `ts_rank(..., 1)` execute inside a single ACID SQL CTE with `status = 'done'` isolation. **Trade-off:** Requires explicit stopword normalization in Python (Phase 4 lifted keyword Recall@6 from `0.29 → 0.86`).
ADR-004 `rag.py • attention.py`	`Local Qwen3-VL-4B + Python Guardrails`	Cloud VLM APIs (GPT-4o) or Prompt-Only Rules	**Why:** 100% air-gapped data privacy and zero API cost; processes both OCR markdown and rendered `896px` page PNGs. **Trade-off:** A 4B model struggles with multi-clause negative prompt exceptions (Phase 3b), so conditional rules (bilingual refusal + citation stripping + `ENOUGH_SCORE ≥ 0.5` gate) are enforced deterministically in Python.
ADR-005 `evaluate.py`	`Human-Calibrated Judge (κ ≥ 0.70 Target)`	Uncalibrated RAGAS / LLM-as-a-Judge	**Why:** Uncalibrated LLM judges hallucinate atomic claims on 1-sentence answers (κ = 0.61 in Phase 2). Deterministic sentence splitting + per-citation verification achieved **κ = 0.79 / 1.00 / 1.00** on the final holdout. **Trade-off:** Required three iterative rounds of 90-row stratified human labeling and SHA-256 judge code tracking (`judge_sha256`).

VERIFICATION ARCHITECTURE

7-Layer Testing, Security & Evaluation Pyramid

Fast unit and database integration tests run locally in under a second with lightweight model stubs (CI workflow pre-configured), while end-to-end security, Playwright browser tests, and golden-set evaluations run against the real local models on Apple Silicon MPS.

Automated Verification Suite — 80 Unit/DB + 25 E2E Pytest Assertions • Isolated *_test Databases

LAYER	SCOPE & ASSERTIONS	EXECUTION TARGET	STATUS
1. Unit Tests (`tests/unit/`)	Text-layer vs. OCR routing, 1,500c table/list re-chunking with repeated headers, `blocked_eager` attention mask, 60-char streaming refusal guardrail, `[n]` parser, 9k token budget, judge sentence deduplication	Local Environment (< 1s, stubbed models; CI-ready)	`80 / 80 PASS`
2. DB Integration (`test_hybrid_search.py`)	Verifies SQL Reciprocal Rank Fusion + stopword-filtered `ts_rank(..., 1)` against exact Python math in real `pgvector:pg17`; ensures half-indexed documents (`status != 'done'`) never shift ranks	Local PostgreSQL 17 Service (Docker / Native)	`PASS (Exact Match)`
3. Prompt Injection E2E (`test_prompt_injection.py`)	Indexes adversarial PDF with hidden white text & fake "Note to AI systems" phishing payload (`PWNED-7731`, `malicious.example`); asserts zero leak in context or answer	Local MPS (Real Models, ~1 min)	`5 / 5 PASS (0 Leaks)`
4. Playwright UI & Refusal E2E (`tests/e2e/`)	Headless Chromium tests for streaming Turkish & English multi-turn queries, rendered page PNGs (`naturalWidth > 0`), search details drawer, Turkish citation-free refusal, and 15/15 refusal judge tests	Local MPS + Chromium (~2.2 min)	`25 / 25 E2E PASS`
5. Retrieval Benchmark (`evaluate.py retrieval`)	Compares 5 search strategies (Vector, Keyword, Hybrid RRF, Vector+Rerank, Hybrid+Rerank) across 69 answerable & 13 unanswerable questions on full 361-page corpus	Local MPS (3 Docs, 1,164 Chunks)	`R@1: 0.88 • R@6: 0.94`
6. Final Holdout Evaluation (`evaluate.py answers`)	72 sealed questions evaluated once without post-hoc tuning: correctness, claim-level faithfulness, citation precision, gold-page citation rate, refusal accuracy, TTFT, and latency	Unseen Holdout (72 Questions)	`0.91 Correct • 0.94 Faith`
7. Judge Calibration (`evaluate.py calibrate`)	Three rounds of stratified 90-row human audits across `claim_supported`, `correct`, and `refusal` using Cohen's κ	Human-in-the-Loop Verification	`κ = 0.79 / 1.00 / 1.00`

BENCHMARKS & ITERATIVE PHASES

Phase-by-phase engineering progression

Every improvement is isolated into a dedicated phase with dated Markdown/JSON reports committed under `phases/faz-N/reports/`. Final success is declared only on a separate, unseen holdout set.

Full Indexed Multimodal Corpus (3 Documents, 361 Pages, 1,164 Chunks)

- **doc1.pdf (ISTQB CT-AI Syllabus):** 99 pages, 298 chunks (max 1,500 chars) — born-digital text layer, risk/acceptance tables, and ML workflow diagrams.
- **doc2.pdf (CODATA 2022 Fundamental Constants):** 62 pages, 304 chunks — dense LaTeX equations and multi-page metrological tables with repeated table headers per split chunk.
- **doc3.pdf (1965 NASA Space Research Report):** 200 full scanned pages, 562 chunks — 100% processed via **Chandra OCR 2** covering optical, solar, radio/radar, X-ray/gamma-ray, and relativistic physics reports.
- **Unseen Final Holdout Set (`eval/golden_final.jsonl`):** 72 stratified questions (10 factual, 10 paraphrase, 10 table, 9 figure, 8 multi-page, 10 Turkish, 15 unanswerable) held sealed until final evaluation.

Critical Engineering Discoveries & Fixes (Phases 1 → 6)

- **MPS Attention & Memory Guard (Phases 2 & J):** PyTorch `sdpa` corrupted long multimodal prompts (`"!!!"`), while standard `eager` hit 50 GB memory spikes and segfaulted on long prompts. Built custom 1024-query `blocked_eager` attention + 9k token budget (`fit_prompt`) + 128-multiple padding (Z13).
- **Why HF `no_repeat_ngram_size` Silently Corrupts RAG (Phase 5b):** Controlled sweeps proved decoder-only n-gram blocking penalizes prompt tokens too, forcing the model to mutate exact source numbers (`2017 → 2020`, `0.5 → 0.7 MeV`). Repetition loops were eliminated instead via 1,500-char table re-chunking (Phase 5).
- **Stopword FTS & Bilingual Refusal Guardrail (Phases 4 & 5c):** Stopword filtering + length-normalized `ts_rank(..., 1)` lifted keyword Recall@6 from `0.29 → 0.86`; normalizing first-sentence refusals in `rag.py` eliminated refusal citation leaks (`0.38 → 0.00`) and enabled native Turkish refusals.
- **Visual Prefill Bottleneck (Phase 6a):** Profiling revealed 3 page PNGs consumed ~3,500 prefill tokens (~66% of prompt); resizing max image side `1280 → 896px` kept 22/22 figure/table accuracy while cutting p50 latency in the 24-question sweep from `32.3s → 20.9s`.

Retrieval Benchmark Progression (Phase 1b/2 Baseline → Phase 4 FTS Fix → Full 361-Page Corpus) — Reports: 20261104-1019 • 20261104-1515 • 20261105-1012 (69 Answerable + 13 Unanswerable Qs)

SEARCH METHOD	PHASE 1B/2 R@1 / R@6 (657 CHUNKS)	PHASE 4 R@1 / R@6 (STOPWORD FTS)	FULL 361P CORPUS RECALL@1 (1,164 CHUNKS)	FULL 361P CORPUS RECALL@6	FULL 361P CORPUS MRR	DOC BREAKDOWN R@6 (DOC1 / DOC2 / DOC3)
Dense Vector (`BAAI/bge-m3`)	0.81 / 0.94	0.81 / 0.94	0.88	0.97	0.914	1.00 / 0.83 / 1.00
Keyword Only (Postgres `TSVECTOR`)	0.07 / 0.29	0.45 / 0.86	0.54	0.86	0.670	0.89 / 0.75 / 0.85
Hybrid RRF (`k = 60`)	0.41 / 0.83	0.64 / 0.88	0.74	0.90	0.817	0.92 / 0.67 / 1.00
Vector + Cross-Encoder Reranker	0.81 / 0.96	0.81 / 0.96	0.88	0.94	0.918	0.97 / 0.75 / 1.00
Hybrid RRF + Reranker (Dev Baseline)	**0.81 / 0.96**	**0.81 / 0.96**	**0.88**	**0.94**	**0.918**	**0.97 / 0.75 / 1.00**
FINAL UNSEEN HOLDOUT (72 Qs)	`SEALED`	`SEALED`	**0.89**	`0.98`	**0.934**	**1.00 / 0.95 / 1.00 (All pages: 1.00)**

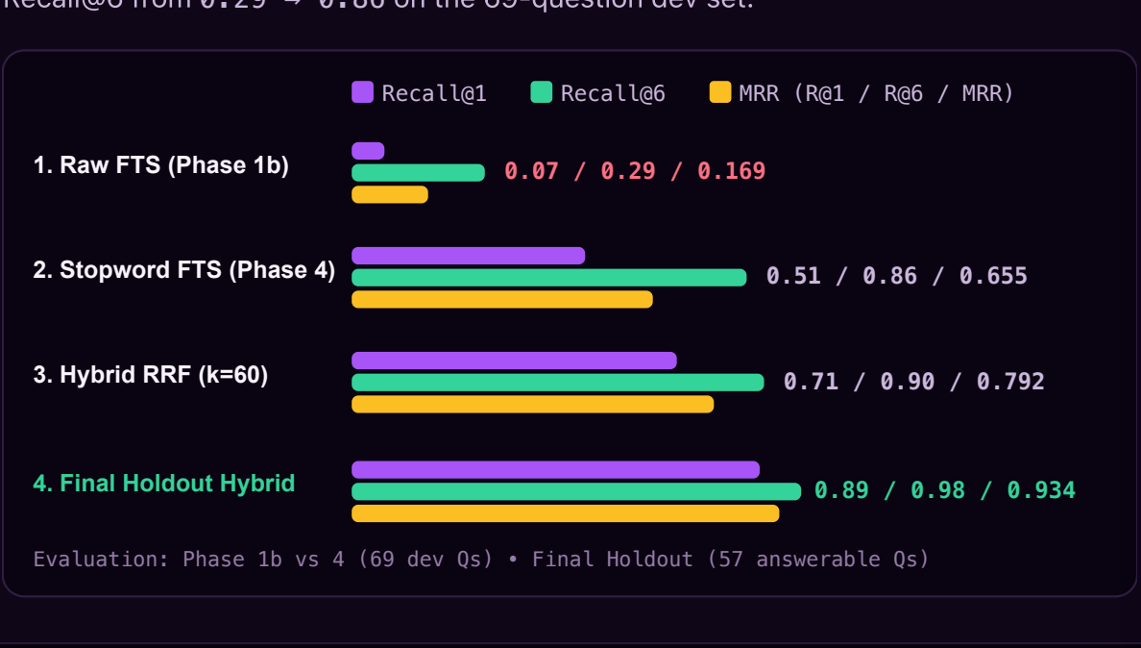
End-to-End Answer Quality, Citation & Latency Progression (Phases 2–8b → Final Unseen Holdout) — Reports: Faz 2 • Faz 5 • Full Baseline • Faz 8b • Final Holdout (20261105-1952)

METRIC	TARGET	PHASE 2 (STABILITY)	PHASE 5 (1.5K RE-CHUNK)	PHASE 8B (2 IMGS, STREAM)	FINAL UNSEEN HOLDOUT (72 QS)	HOLDOUT STATUS
Completed Questions	100%	82 / 82	82 / 82	82 / 82	**72 / 72** (57 Ans + 15 Refusal)	`100% (0 Crashes)`
Answer Correctness	≥ 0.85	0.91	0.97	0.97	**0.91** (52 / 57 correct)	`0.91 ( PASS )`
Faithfulness (Judge)	≥ 0.92	0.93	0.94	0.96* (Verified)	**0.94** (Raw judge score)	`0.94 ( PASS )`
Citation Precision	≥ 0.80	0.83	0.84	0.84	**0.87**	`0.87 ( PASS )`
Cites Expected Gold Page	≥ 0.85	`0.62`	0.90	0.90	**0.95**	`0.95 ( PASS )`
Answers Without Citation	≤ 0.05	`0.30`	0.00	0.00	**0.00**	`0.00 ( PASS )`
Unanswerable Refused	≥ 0.90	0.92 (12/13)	1.00 (13/13)	1.00 (13/13)	**1.00** (15 / 15)	`1.00 ( PASS )`
Unanswerable With Citation	0.00	`0.08`	`0.15`	0.00	**0.00** (Raw: 0.07*)	`0.00 ( PASS )`
False Refusals	≤ 0.05	0.00	0.00	0.00	**0.02** (1 / 57, f51)	`0.02 ( PASS )`
Latency (p50 / p95)	≤ 30s / ≤ 60s	`49.9s / 97.8s`	35.4s / 67.0s	17.8s / 52.4s	**22.2s / 40.4s**	`22.2s ( PASS )`
Streaming TTFT (p50)	≤ 15s (Rev. from ≤10s)	—	—	12.9s	**15.2s** (0.2s over target)	`ACCEPTED (Owner decision)`

`ABLATION STUDY 01 • RETRIEVAL PIPELINE`

Retrieval Ablation: Sparse FTS Fix → Hybrid RRF → Cross-Encoder

Removing English/Turkish stopwords and normalizing document length (`ts_rank(..., 1)` in Phase 4) lifted keyword Recall@1 from `0.07 → 0.51` (7.3×) and Recall@6 from `0.29 → 0.86` on the 69-question dev set.

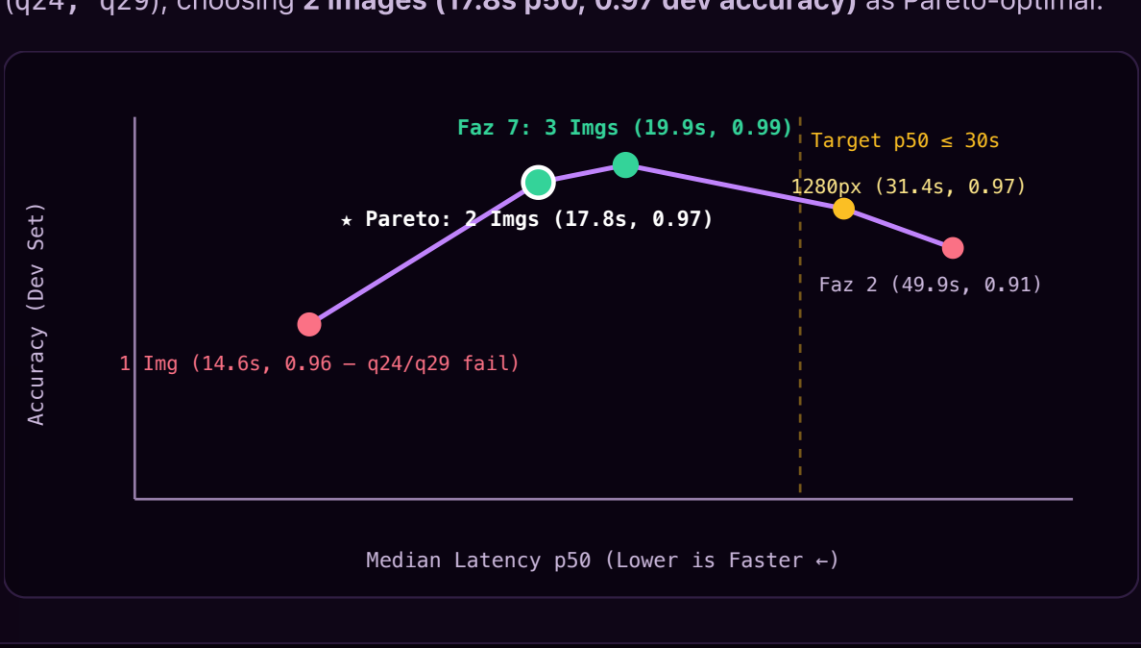


Evaluation: Phase 1b vs 4 (69 dev Qs) • Final Holdout (57 answerable Qs)

`ABLATION STUDY 02 • LATENCY vs. ACCURACY PARETO`

Visual Token Prefill & Page Image Count Pareto Analysis

Downscaling page PNGs from `1280px → 896px` halved visual prefill tokens. Reducing `MAX_PAGE_IMAGES` from 3 → 1 cut TTFT to `10.8s` but regressed on 2 questions (`q24, q29`), choosing **2 images (17.8s p50, 0.97 dev accuracy)** as Pareto-optimal.



FAILURE ANALYSIS & POST-MORTEMS

What failed during development & how we proved the root cause

Instead of cherry-picking only successful runs, every failed hypothesis or regression was isolated, root-caused with concrete question IDs, and either rolled back or solved at the architectural level.

`POST-MORTEM 01 • PHASE 5b` `ROLLED BACK & FIXED IN CHUNKER`

Why Hugging Face `no_repeat_ngram_size` Silently Corrupts RAG Facts

Symptom: On dense tables (`q33, q36, q37, d2q10`), the 4B VLM occasionally entered a sentence repetition loop.

Failed Hypothesis: Adding standard `no_repeat_ngram_size=5` or `repetition_penalty=1.05` during generation.

Root Cause Discovered: Controlled sweeps (`phases/faz-5b/RAPOR.md`) proved HF's n-gram filter blocks n-grams appearing in the *input prompt* too. When a question and source excerpt shared a 5-gram, the model was forbidden from repeating the exact year or number, silently mutating `2017 → 2020` (q01) and `0.5 MeV → 0.7 MeV` (d2q05).

Engineering Resolution: Rejected all decoding penalties. Eliminated loops at the root by splitting 6,104-char tables into `≤1,500`-char chunks with repeated headers (Phase 5) and deduplicating identical sentences in the judge.

`POST-MORTEM 02 • PHASES 2 & J (Z9–Z13)` `SOLVED VIA BLOCKED_EAGER`

Multimodal Long-Context `sdpa` Collapse vs. 50 GB `eager` Memory Spikes

Symptom: On q43 (13.3k input tokens with 3 high-res page PNGs), standard `eager` attention caused an MPS segfault/OOM (spiking up to 50 GB memory footprint on `d2q12`).

Failed Hypothesis: Switching Qwen3-VL to PyTorch `sdpa` (Scaled Dot-Product Attention). Above ~9,500 multimodal tokens, `sdpa` silently overflowed and emitted `"!!!!!!"` garbage while corrupting subsequent calls in the same process.

Engineering Resolution: Implemented custom `blocked_eager` attention in `attention.py` (computing the $Q \cdot K^T$ matrix in 1,024-query blocks), enforced a 9,000-token budget (`rag.fit_prompt`), and padded embeddings/reranker inputs to multiples of 128 (Z13) to stop MPS graph cache growth (cutting peak memory from 50 GB to ~9 GB).

`POST-MORTEM 03 • PHASE 8` `CAUGHT BY NON-REGRESSION GATE`

Single-Image Context Starvation & TOC Loop on q29 and q24

Symptom: To push streaming TTFT under 10s, we tested reducing `MAX_PAGE_IMAGES` from 3 → 1. On the 22-question figure/table sweep (`sweep8.json`), 1 image scored a perfect 22/22 with `10.6s` TTFT.

Full-Set Regression: On the full 82-question dev benchmark, 1 image achieved `10.8s` TTFT and `0.93` raw faithfulness, but regressed on two questions (`0.99 → 0.96`): on q29, missing the second page image caused the VLM to loop over the Table of Contents and cite non-existent chunks `[7][8]`; on q24, it hallucinated *"test cases or inputs"*.

Engineering Resolution: Enforced our strict non-regression rule (max 1 question drop allowed for speed) and selected `MAX_PAGE_IMAGES = 2`, which restores q29 while cutting TTFT from 17.7s to 12.9s and overall p50 latency from 19.9s to 17.8s.

`POST-MORTEM 04 • PHASES 3b, 5c & 6a` `SOLVED IN PYTHON GUARDRAIL`

Conditional Prompt Exceptions in 4B Models & Cross-Chunk Attribution Drift

Symptom 1 (Refusal Citations): Phase 3's mandatory citation prompt reduced uncited answers from `0.30 → 0.00`, but caused the model to append `[1]` to refusal sentences (`0.38`). Phase 3b's prompt-only exception (*"Cite every sentence EXCEPT when refusing"*) had zero effect (`0.38 → 0.38`).

Symptom 2 (Attribution Drift): Human audit of all 16 "unsupported" claims in Phase 6a revealed **7/16 were judge false negatives** (facts were explicitly in the chunk, lifting true faithfulness to **0.95**), while human inspection of the remainder showed most unsupported flags were accurate facts cited to adjacent page chunks.

Engineering Resolution: Moved conditional refusal logic out of the prompt into a deterministic 60-character streaming buffer in `rag.py` (`is_refusal` → clean bilingual refusal with `0.00` citations).

HUMAN-IN-THE-LOOP CALIBRATION

LLM-as-a-Judge audit on stratified decisions across Dev & Final Holdout

Because answers are evaluated locally by the same 4B vision-language model (so no private document data leaves the machine), we audited three rounds of 90 stratified judge decisions against human ground-truth labels using Cohen's κ.

Stratified Human vs. Local 4B Judge Agreement (Phase 2 → Phase J → Final Holdout) — Reports: 20261104-1330 • 20261104-2303 • Final Holdout 20261105-1952 (3 × 90 Human Labels)

JUDGMENT KIND	PHASE 2 (AUDIT)	PHASE J (STRATIFIED RE-AUDIT)	FINAL HOLDOUT (72 QS AUDIT)	TARGET & STATUS
`refusal`	κ = 1.00	13 positive refusals + 17 non-refusals **1.00 (κ = 1.00)**	16 positive refusals + 14 non-refusals **1.00 (κ = 1.00)**	`κ ≥ 0.70 (PERFECT)`
`correct`	κ = 0.76	4 negative verdicts + 26 positive verdicts **0.93 (κ = 0.76)**	5 negative verdicts + 25 positive verdicts **1.00 (κ = 1.00)**	`κ ≥ 0.70 (PERFECT)`
`claim_supported`	`κ = 0.61`	17 negative verdicts + 13 positive verdicts **0.93 (κ = 0.86)**	7 negative verdicts + 23 positive verdicts **0.93 (κ = 0.79)**	`κ ≥ 0.70 (PASS)`

SYSTEM LIMITATIONS & HONEST OBSERVATIONS

Where the system currently falls short & open engineering challenges

True engineering maturity means acknowledging boundary conditions and failure modes transparently rather than claiming flawless production readiness.

1. Multilingual Performance Gap (Turkish vs. English)

On the final unseen holdout, Turkish retrieval remained robust at **Recall@6 = 0.90**, yet answer correctness dropped to **0.70** (vs. 0.95+ on English questions), isolating the primary bottleneck to generation rather than search.

- **Language-Drift in Generation (Resolved in Phase 9):** The 4B VLM occasionally answered Turkish queries in English (e.g., `f57`). Phase 9 introduced deterministic `answer_language()` enforcement, reducing language mismatch to `0` with 89 passing tests.
- **Cross-Lingual Terminology Mismatch (Tested in Phase 10):** Conceptual Turkish queries (e.g., *"Kavram kayması nedir?"*) cannot match English passages without query translation. Phase 10 proved translation lifted dev correctness to **0.99** (12/12 Turkish correct), but added ~9s latency; rolled back to Phase 9 to respect TTFT limits. Phase 10 code preserved; single-pass reranking is the planned fix.
- **Inflectional Stemming Deficit:** PostgreSQL full-text search uses the unstemmed `simple` dictionary, so inflected Turkish words rely on dense vector recall.
- **Failed Holdout Questions:** `f49` (flexibility vs. resilience nuance), `f51` (over-cautious false refusal on valid passage), and `f54` (omitted binding energy derivation).
- **Governance Artifacts:** Full system reporting committed in `docs/MODEL_CARD.md` and risk self-assessment in `docs/NIST_AI_RMF.md`.

2. Hardware & Local VLM Constraints

Executing completely air-gapped on local unified memory entails physical trade-offs against multi-billion parameter cloud APIs:

- **Streaming TTFT (15.2s vs. 15s Target):** Processing two 896px page images plus cross-encoder reranking incurs ≈4–5 s retrieval + ≈7–10 s prefill before the first generated token appears.
- **4B Parameter Ceiling:** 4B models struggle with complex compound instructions (e.g., `f39` missed the specific bolometer instrument and `f44` lost multi-page proton radius nuances).
- **Local Single-User Architecture:** Current database connections and model memory footprints are optimized for a dedicated workstation, requiring batch queueing for multi-tenant deployment.

MULTIMODAL UI & SOURCE VERIFICATION

Interactive Streamlit interface & real indexed page sources

Every answer displays clickable page thumbnails and an expandable **Search details** drawer exposing the rewritten queries, hybrid retrieval candidates, and Cross-Encoder rerank scores.

Streamlit App (app.py) — Illustrative UI Walkthrough

Data Source

✓ Indexed Documents (3 • 361 Pages)

- `doc1.pdf (99p, 298 chunks)`
- `doc2.pdf (62p, 304 chunks)`
- `doc3.pdf (200p, 562 chunks)`

User: In the ML workflow figure, which steps come after testing the model, and what are the two mirror shapes in the grazing incidence X-ray telescope?

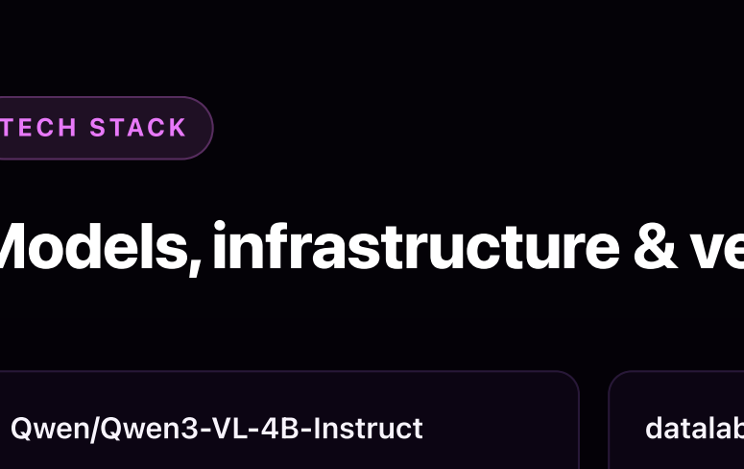
Assistant: In the ML workflow diagram, after *Test the model*, the workflow proceeds to **Deploy the model**, **Use the model**, and finally **Monitor and tune the model** [1]. In the grazing incidence X-ray telescope schematic, the two mirror shapes used are a **paraboloid** and a **hyperboloid** [2].

`▸ Search details: Queries: `ML workflow figure steps after testing model` → `grazing incidence x-ray telescope mirror shapes` • Top Rerank Scores: 0.996, 0.994`

`doc1.pdf • Page 29 (Figure)`

Born-Digital Diagram Grounding

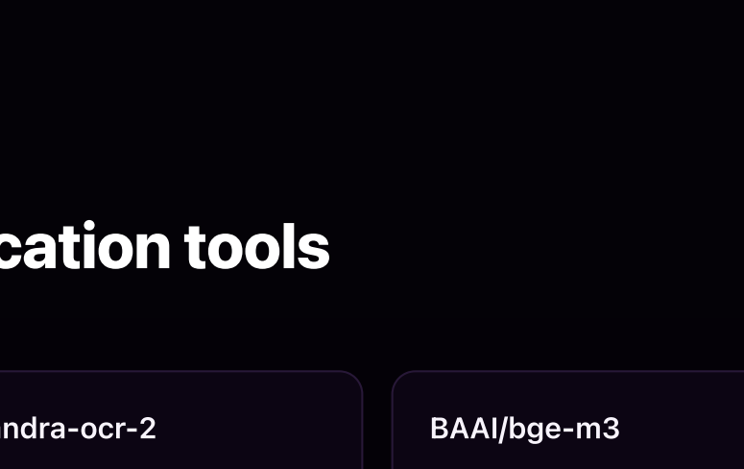
ISTQB CT-AI ML workflow diagram extracted via Docling + passed as a visual page image to Qwen3-VL.



`doc2.pdf • Page 45 (Table XXXIII)`

Dense Metrological Table Extraction

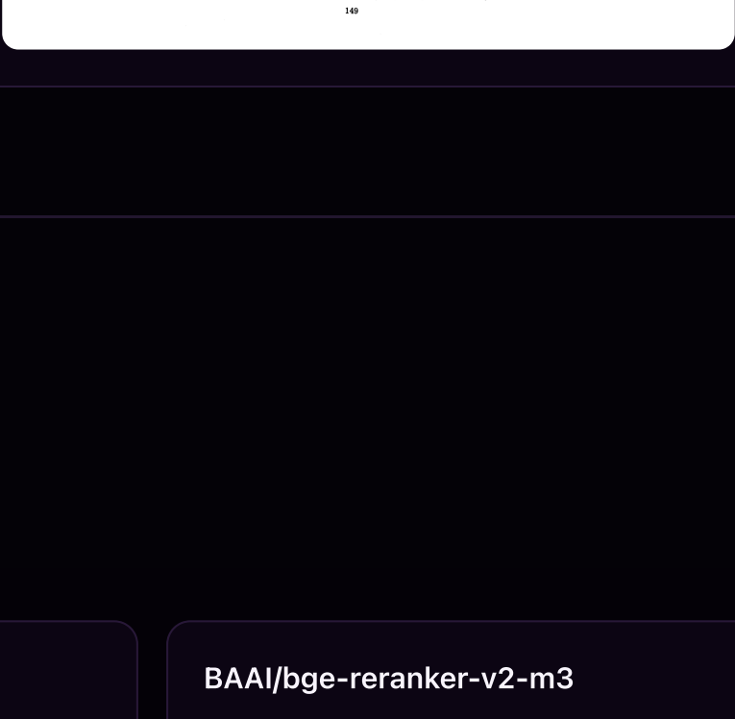
CODATA 2022 fundamental physical constants table preserving exact uncertainties and exponent notation.



`doc3.pdf • Page 157 (Sample Page 33)`

1966 Scanned NASA Report (Chandra OCR)

Full-page scan routed to `chandra-ocr-2` (original report page 157) showing grazing incidence X-ray optics.



TECH STACK

Models, infrastructure & verification tools

Qwen/Qwen3-VL-4B-Instruct	Multimodal VLM Generator
datalab-to/chandra-ocr-2	Layout-Aware VLM OCR
BAAI/bge-m3	1024-d Dense Embeddings
BAAI/bge-reranker-v2-m3	Cross-Encoder Reranker
PostgreSQL 17 + pgvector	HNSW Vector + GIN FTS RRF
Docling + pypdfium2	PDF Layout & Image Coverage
Pytest + Playwright	Unit, DB, Injection & UI E2E
Streamlit + PyTorch MPS	Local Interactive UI & Runtime

Built as an engineering-grade Multimodal Agentic RAG portfolio project with hypothesis-driven phases and calibrated evaluation.

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